

DATA VORTEX

Round 1 – Phase 1 Report

Rebuilding the Social Engine

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Data Recovery • Cleaning • Validation • Exploratory Data Analysis

1. Executive Summary

DATA VORTEX Round 1 – Phase 1 focuses on rebuilding a deliberately corrupted social-media dataset. The objective was to recover the available data, restore data quality, identify anomalies, validate consistency, and perform exploratory analysis without fabricating information.

The recovered Social Engine dataset contains two related tables: a users table with 1,500 users and a posts table that contained 12,360 rows before cleaning. The posts data contained duplicate records, missing platform values, missing and corrupted text, mixed timestamp formats, and invalid negative like counts.

The final cleaned posts dataset contains 12,000 unique posts and the users dataset contains 1,500 users. All missing values were handled, negative engagement values were resolved, timestamps were standardized, corrupted text was normalized, and referential integrity between posts and users was verified.

EDA indicates broadly stable posting activity and engagement over the observed period. Platform-level average engagement is relatively close across platforms. Text length is concentrated in a narrow range, while follower count shows almost no linear relationship with engagement or posting frequency.

2. Dataset Overview

The users table contains user_id, location, language, account_created, and follower_count. The posts table contains post_id, user_id, platform, text_content, timestamp, likes, shares, and comments.

Dataset	Rows	Columns	Purpose
Users	1,500	5	User attributes and follower information
Posts (raw)	12,360	8	Social-media posts and engagement metrics
Posts (cleaned)	12,000	8	Validated analytical dataset

3. Data Recovery and Cleaning

The corrupted dataset was recovered from the Social Engine challenge environment. The recovered files were processed through a reproducible pandas-based cleaning pipeline. Each transformation was selected to correct an observed data-quality issue while preserving valid records.

Issue	Observed Problem	Treatment
Duplicate posts	12,360 raw rows vs. 12,000 unique post IDs	Deduplicated by post_id and retained the first record
Missing platform	1,846 missing platform values	Filled with 'Unknown'
Missing/corrupt text	NaN, literal NULL and corrupted NULL variants	Standardized to [MISSING_TEXT]
HTML corruption	HTML tags and encoded entities	Removed tags and decoded HTML entities
Mojibake	Corrupted UTF-8 representations such as Å	Applied safe latin1/UTF-8 repair where valid
Likes	525 negative values and 1,858 missing values	Invalid/missing likes imputed using platform-aware medians; overall valid median for Unknown
Timestamps	ISO, Unix epoch and DD-MM-YYYY formats	Parsed and standardized into datetime values
User references	Posts linked to users through user_id	Validated against users table

3.1 Text Cleaning

Text cleaning removed HTML tags, decoded HTML entities, attempted safe repair of mojibake, and converted missing or NULL-like text to the explicit placeholder [MISSING_TEXT]. After cleaning, no NULL-like text values remained. There were 1,779 posts represented by [MISSING_TEXT], accounting for the original missing and NULL-like text cases.

3.2 Engagement Cleaning

Likes were the only engagement metric containing negative values. Missing and invalid likes were treated as missing and imputed using median values so that the correction would be robust to extreme values. Shares and comments did not contain negative values and were retained.

3.3 Timestamp Standardization

All timestamps were successfully parsed. The cleaned dataset spans from May 1, 2024 to April 30, 2025. A limitation was identified during temporal EDA: date-only DD-MM-YYYY records naturally parse to midnight, so hour-of-day patterns should not be interpreted as genuine user posting behavior.

4. Data Validation

These checks confirm that the cleaned data is structurally consistent and suitable for downstream analysis. The final exported files preserve the intended analytical fields and do not include temporary EDA-only columns such as total_engagement.

Validation Check	Result
Final posts	12,000 rows × 8 columns
Final users	1,500 rows × 5 columns
Missing values in final datasets	0
Duplicate post IDs	0
Negative likes	0
Negative shares	0
Negative comments	0
Invalid post → user references	0
Earliest post timestamp	2024-05-01 00:00:00
Latest post timestamp	2025-04-30 21:57:10

5. Exploratory Data Analysis

5.1 Platform Distribution and Engagement

After cleaning, platform counts are reasonably balanced among the five named platforms, with an additional Unknown category representing records where the original platform value was missing. Facebook and YouTube have just over 2,000 posts each, while Instagram has slightly fewer. Unknown contains approximately 1,784 posts.

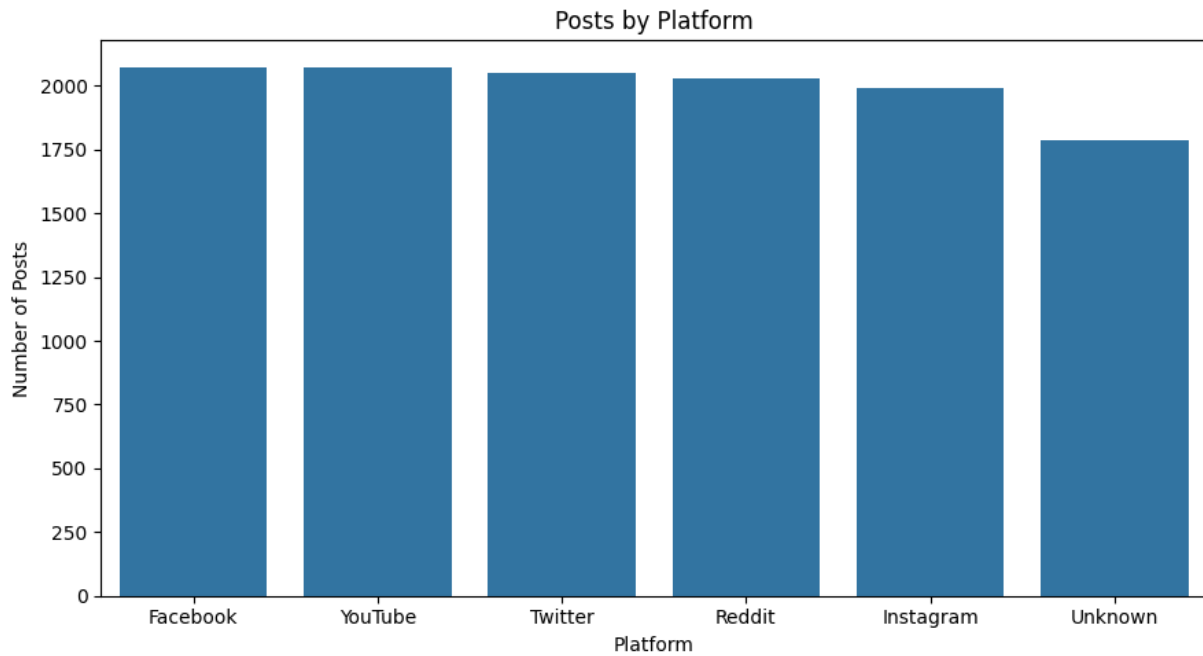


Figure 1. Distribution of cleaned posts across platforms.

Average total engagement is also relatively close across platforms. The highest average in the supplied chart is for YouTube, while Twitter is slightly lower. The relatively narrow range suggests that platform alone does not explain large differences in average engagement in this dataset.

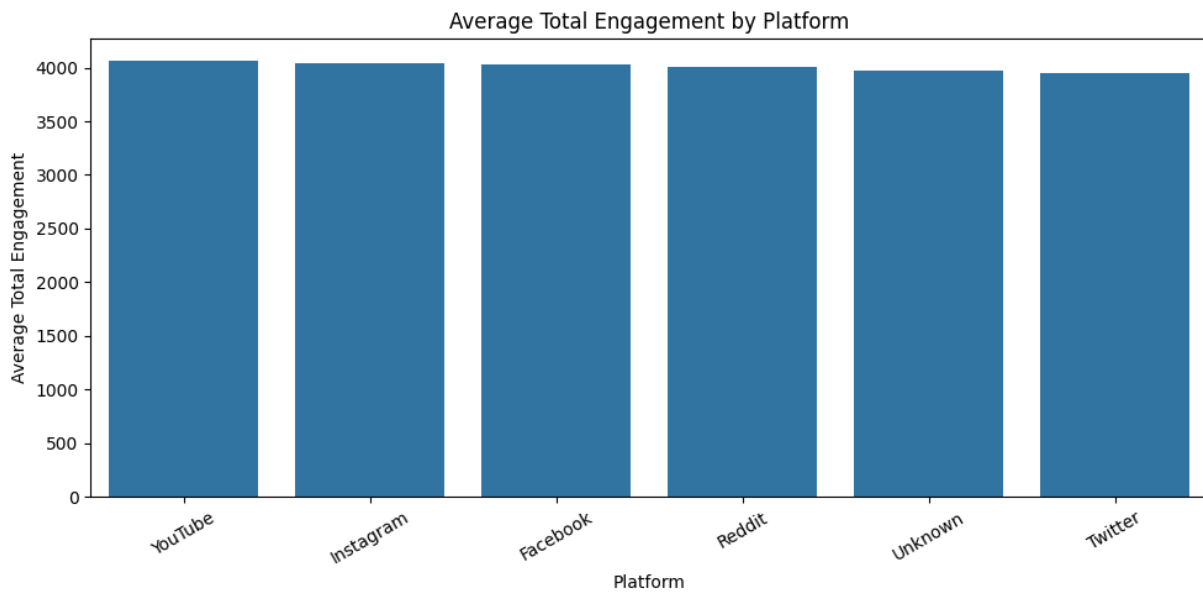


Figure 2. Average total engagement by platform.

5.2 Engagement Distribution

Total engagement was calculated temporarily as likes + shares + comments for EDA. The distribution is centered around roughly 4,000 total interactions per post and shows a broadly bell-shaped pattern. This supports mean and median comparisons while still motivating explicit anomaly checks.

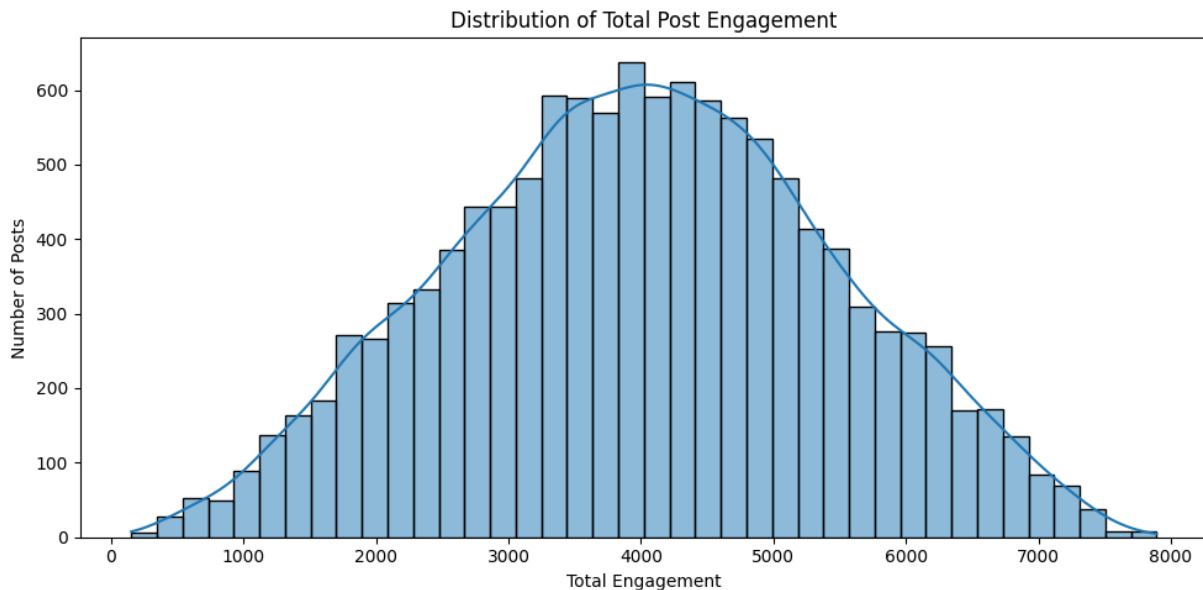


Figure 7. Distribution of total post engagement.

5.3 Temporal Analysis

Monthly posting activity remained broadly stable from May 2024 through April 2025. The highest monthly post count was 1,038 in May 2024, while the lowest was 914 in February 2025. The February decrease is modest in the context of the full year and was not classified as an anomaly.

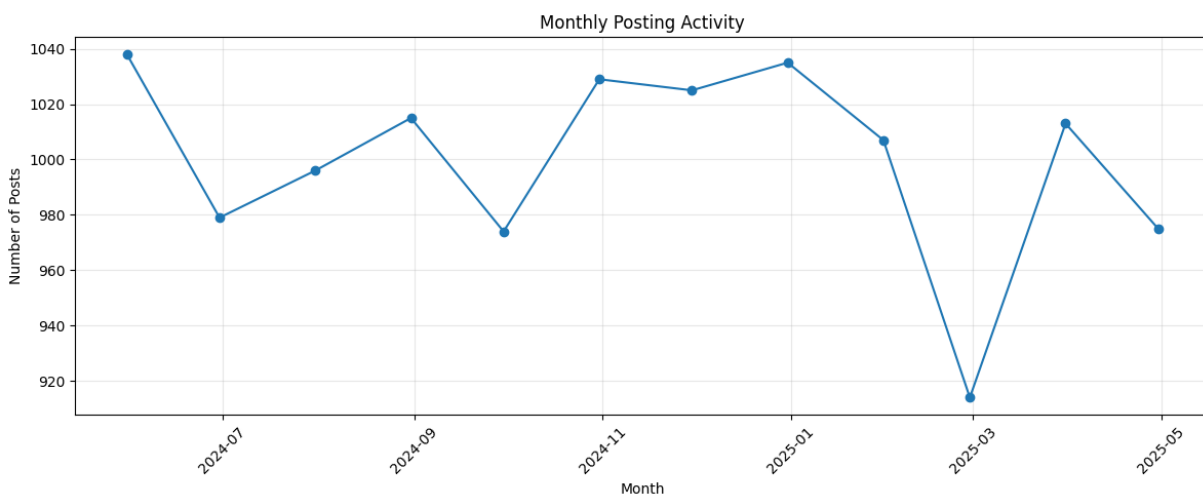


Figure 3. Monthly posting activity.

Posting volume is also distributed fairly evenly across weekdays. Wednesday has the highest count at 1,771 posts, while Saturday has the lowest at 1,675. The difference of 96 posts is small relative to the total dataset, indicating no strong day-of-week concentration.

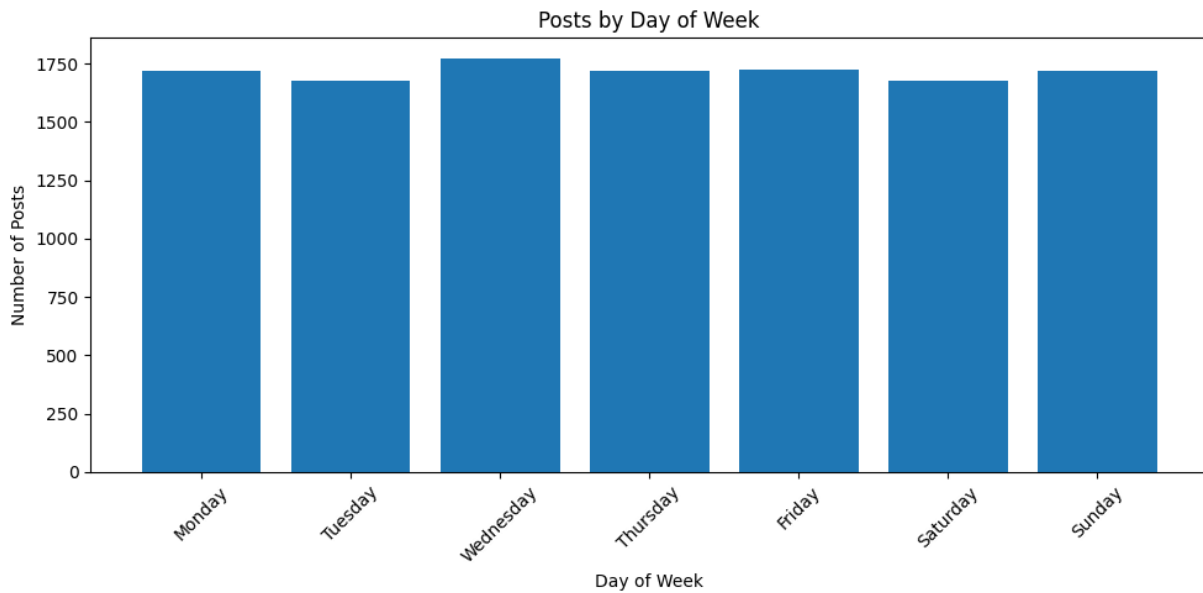


Figure 4. Posts by day of week.

Monthly average likes, shares, and comments remain relatively stable throughout the period. Average likes fluctuate around the mid-2,000s, average shares around 1,000, and average comments around 500, with no sustained upward or downward trend.

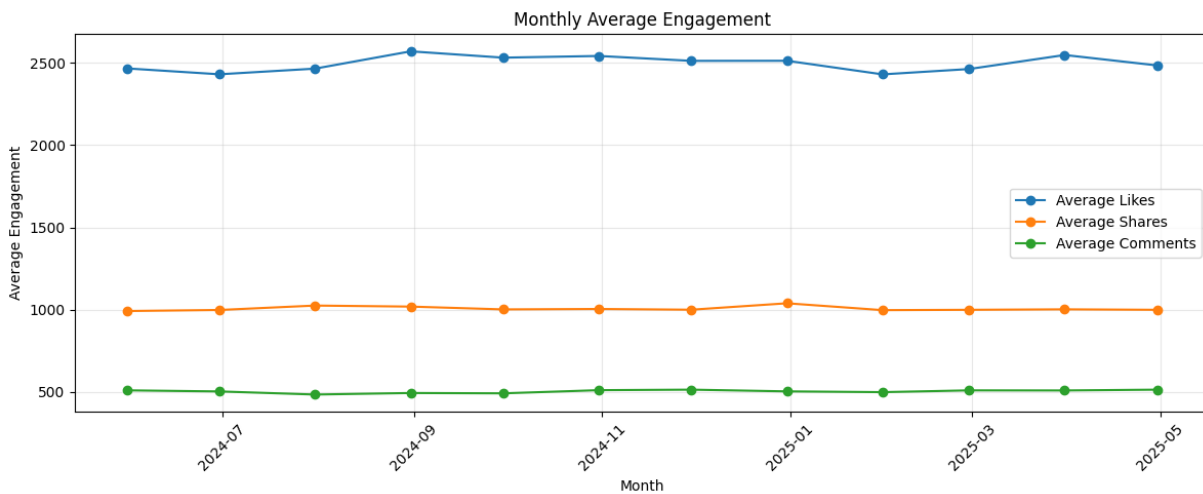


Figure 5. Monthly average engagement components.

Hour-of-day analysis was intentionally not used for behavioral interpretation because date-only timestamps were parsed at midnight. Treating those midnight values as actual midnight posting behavior would introduce a misleading conclusion.

5.4 Content Analysis

Among 12,000 cleaned posts, 10,221 contain valid text content. Text length has a tight distribution: mean 117.87 characters, median 118, Q1 107, Q3 129, and maximum 172 characters. The majority of valid posts therefore fall within a relatively narrow length range.

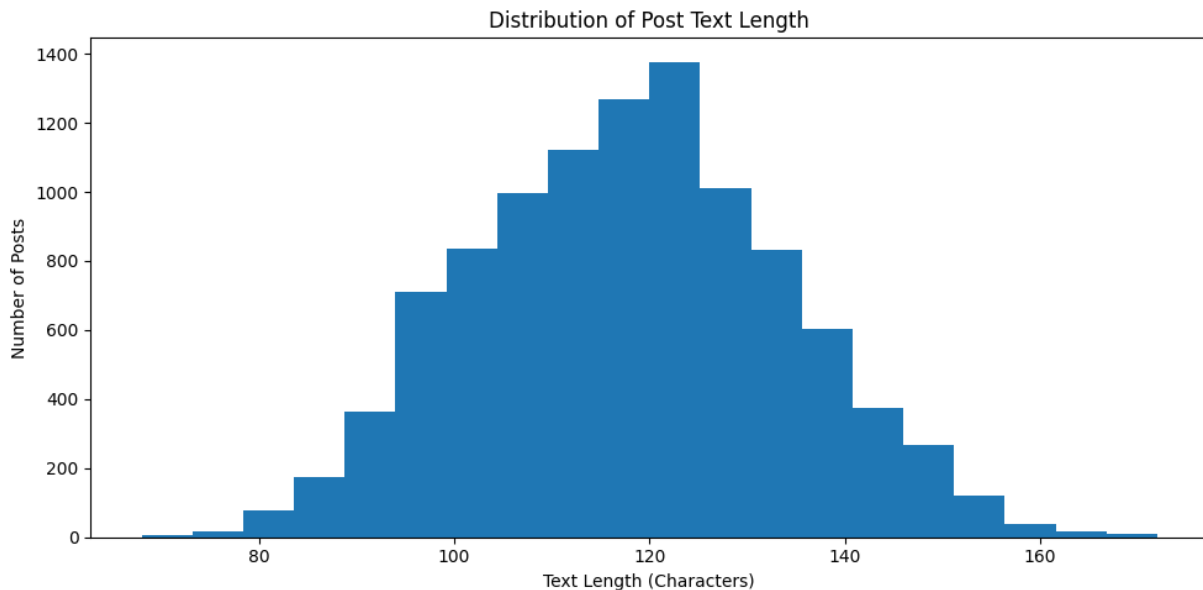


Figure 6. Distribution of valid post text length.

Length categories show that 8,497 posts (83.1%) contain 101–150 characters, 1,516 posts are between 51–100 characters, and 208 are between 151–200 characters. No valid-text posts fall below 51 or above 200 characters under the selected bins.

Linear correlations between text length and engagement were close to zero: likes -0.008 , shares -0.028 , and comments -0.015 . Within this dataset, text length therefore does not appear to be a meaningful linear predictor of engagement.

5.5 Hashtags, Mentions and Content Patterns

Every valid text post contains at least one hashtag. Hashtag counts are concentrated between one and three hashtags: 3,358 posts contain one, 3,416 contain two, and 3,447 contain three.

Mentions are less common. 8,845 valid posts contain no mentions, 679 contain one, and 697 contain two, meaning approximately 13.46% of valid-text posts contain at least one mention.

Posts without mentions average approximately 2,504.53 likes, 1,004.21 shares, and 502.62 comments. Posts with mentions average approximately 2,417.35 likes, 1,009.82 shares, and 515.37 comments. These differences are small and should not be interpreted as causal effects.

Common-word analysis is dominated by feedback-oriented language and brand/product discussion. Frequently occurring terms include 'feedback', 'experience', 'quality', 'recommend', 'review', and brand-related terms such as Pepsi, Coca-Cola, Apple, and Adidas. This suggests that a substantial portion of the dataset is oriented toward opinions, experiences, and product feedback.

5.6 User and Follower Analysis

Posting activity is broadly distributed across the 1,500 users. The average and median number of posts per user are both 8, with a standard deviation of 2.85. The minimum is 1 and the maximum is 22 posts per user. This indicates moderate variation without extreme concentration in a small group of users.

Follower counts range from 109 to 49,944, with a mean of 24,964 and a median of 24,741.5. Despite this variation, follower count has essentially no linear relationship with engagement: correlations are 0.008 with likes, -0.014 with shares, and approximately 0 with comments.

Posting frequency is also almost identical across follower quartiles. Average posts per user range only from about 7.87 to 8.12 across the four quartiles. The correlation between follower count and post count is -0.008.

6. Anomaly Analysis

Anomaly detection combined statistical rules with data-integrity checks. The goal was to distinguish genuine unusual values from corruption introduced into the dataset.

Check	Result	Decision
Likes IQR outliers	0	No records removed
Shares IQR outliers	0	No records removed
Comments IQR outliers	0	No records removed
Zero likes	1	Retained as valid
Zero shares	5	Retained as valid
Zero comments	13	Retained as valid
Duplicate post IDs after cleaning	0	Passed
Invalid user references	0	Passed

IQR thresholds were calculated separately for likes, shares, and comments. The upper bounds were 6,304.5 for likes, 2,987.5 for shares, and 1,508 for comments; no observations exceeded these thresholds. Therefore, high engagement values were not removed merely because they were large.

Some posts have shares greater than likes or comments greater than likes. These relationships were not treated as anomalies because there is no valid data rule requiring likes to be greater than the other engagement measures.

7. Key Insights

1. The raw posts table contained 360 excess duplicate rows; deduplication produced 12,000 unique posts.
2. The cleaned dataset contains no missing values, duplicate post IDs, negative engagement values, or invalid user references.
3. Platform distribution is reasonably balanced, although 1,784 posts remain categorized as Unknown because the original platform value was missing.
4. Average engagement is relatively similar across platforms, indicating no strong platform-level separation in this dataset.
5. Posting activity is stable over the year, with May 2024 highest at 1,038 posts and February 2025 lowest at 914.
6. Wednesday is the most active day with 1,771 posts, while Saturday is the least active with 1,675.
7. Valid post text has a mean length of 117.87 characters and a median of 118; 83.1% falls in the 101–150 character range.
8. Text length has near-zero linear correlation with likes, shares, and comments.
9. Follower count has near-zero correlation with engagement and posting frequency.
10. The content is strongly feedback-oriented, with frequent product/brand and experience-related vocabulary.
11. No IQR-based engagement outliers were found, so no legitimate high-engagement posts were removed.

8. Reproducibility and Outputs

The Phase 1 workflow is implemented in a Jupyter notebook using Python and pandas. The workflow follows a consistent sequence: load recovered data → inspect quality → clean and standardize → validate → derive temporary EDA features → analyze → export cleaned datasets.

Output	Description
Social_Engine_Users_Cleaned.csv	Validated users table with 1,500 rows
Social_Engine_Posts_Cleaned.csv	Validated posts table with 12,000 rows
Jupyter Notebook	Cleaning, validation, EDA and reproducible workflow
Phase 1 Report	Documentation of methodology, findings and insights

9. Conclusion

The Social Engine dataset was successfully recovered and rebuilt for Phase 1 of DATA VORTEX Round 1. The cleaning process addressed duplicate records, missing platform values, corrupted text, mixed timestamp formats, and invalid or missing likes while preserving valid observations.

The final datasets passed structural and referential validation. EDA shows a stable social-media environment with broadly consistent posting activity, relatively similar platform-level engagement, concentrated text lengths, and weak relationships between follower count, text length, posting frequency, and engagement.

The analysis also demonstrates the importance of distinguishing corruption from legitimate variation. No IQR-based engagement outliers were found, and unusual relationships between likes, shares, and comments were not removed without a domain-based rule. This preserves the analytical integrity of the recovered dataset.

Appendix A – Cleaning and Validation Methodology

The cleaning pipeline uses explicit transformations rather than silently discarding records. Text normalization converts missing or NULL-like text to [MISSING_TEXT], HTML is stripped and decoded, and mojibake is repaired only when the conversion succeeds. Missing platforms are represented as Unknown. Missing or invalid likes are imputed using median-based logic. Timestamps are normalized to a consistent datetime representation.

Before final export, temporary analysis columns such as total_engagement and text_length are removed so that the final posts file retains the intended eight source fields. Final checks confirm zero missing values, zero duplicate post IDs, zero negative engagement values, and zero invalid user references.